Text as Pieces

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Text as Pieces

> Prepare Text as Pieces: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Encode and decode text under an explicit toy **vocabulary**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Feedback Loops

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

Look for: Implement a deterministic recommender

Reference example, not the only acceptable wording: Both score 8 and qualify; A is alphabetically first under the tie rule.

Nearest Neighbor Reasoning

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and rank **training** neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Define a question and target population

Reference example, not the only acceptable wording: Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Toy **token**izer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count **token**s for each and explain why a three-word sentence need not use three **token**s.

Concept to explain

Language systems **process token**s and **pattern**s rather than whole human meanings.

Term	Meaning
token	A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation.
tokenization	Converting text into the token units of a particular tokenizer.
context	Information available around a task that helps interpret or answer it.
pattern	A relationship or regularity observed in examples.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

The preceding **token** sequence, together with the model and any stated **context**. Changing segmentation changes that **token** sequence.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

End goal for the pictorial case

By the end, I can encode and decode text using a specified **vocabulary** and matching rule.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Text as Pieces using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. ‘unhelpful’ might split as ‘un’ + ‘help’ + ‘ful’.

2. A name or emoji may use unexpected **token** pieces.

3. **Token** boundaries need not match whole words.

4. Meaning emerges from **pattern**s across **context**, not from one **token** acting like a dictionary.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

The teaching case

Count **token**s and state the limit of the example.

Exact explanation to model

Why the result follows

Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

Explain the added worked example

Explain the added worked example · 1

Tokens need not be words. A classroom **token**izer can use an exact, case-sensitive **vocabulary** and greedily choose the longest matching **token** from left to right. This is a toy rule, not a claim about every production **token**izer.

Define IDs: 0=end, 1=cat, 2=c, 3=a, 4=t, 5=s, 6=space, 7=sat. The space **token** represents one literal space. The end **token** is a special terminator, not ordinary text.

Explain the added worked example · 2

At each position choose the longest matching ordinary **token**. If no **token** matches, stop and report the unsupported character and its position. Do not delete the character or guess a replacement.

IDs identify entries; a larger ID does not mean a **token** is more valuable or more similar.

Explain the added worked example · 3

The **encoding** of cats sat is [1,5,6,7], followed by [0] only if the protocol appends an end **token**. **Decoding** concatenates the ordinary **token** strings to recover cats sat.

Count four text **token**s, or five sequence items with the end marker included; always state the counting convention. The two written words do not imply two **token**s.

Key terms

Term	Meaning
Token	An item in a model vocabulary, which may represent a word, part of a word or another text unit
Vocabulary	The permitted tokens and their identifiers
Encoding	Mapping source text into a token sequence under a specified rule
Decoding	Mapping a token sequence back to text under a specified rule

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 4

A character-only **vocabulary** c,a,t,s,space could encode cats sat as c,a,t,s,space,s,a,t: eight text **token**s. The original **vocabulary** uses four. Both decode to the same source text.

Sequence length is a property of the text, **vocabulary** and **encoding** rule together. Counts from different **token**izers are not automatically comparable.

In this **vocabulary**, sat is chosen over s because it is longer. The toy longest-match rule is defined explicitly. Real **token**izers may use different algorithms, byte **encoding**s and special-**token** policies.

Never infer an unknown **token**izer's count from word count alone.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Text as Pieces

A fictional **token**izer converts a short catalogue phrase into **token**s.

The facts we will use

> **Supplied facts**
>
> **Vocabulary** entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

> **Supplied facts**
>
> Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.

AI ACADEMY / AI-2/25

Text as Pieces

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read prefix

At position 0, longest matching entry is books, not book.



2. Encode phrase

Tokens: books, space-character, book; three tokens.



3. Decode

Joining token text returns exactly books book.



4. Unknown character

For books book!, the ! at position 10 has no token: stop with error.

CHANGE ONE THING

Use a character vocabulary covering b,o,k,s and space instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read prefix	At position 0, longest matching entry is books, not book.
2. Encode phrase	Tokens: books, space-character, book; three tokens.
3. Decode	Joining token text returns exactly books book.
4. Unknown character	For books book!, the ! at position 10 has no token: stop with error.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly.

What this example does not establish

Keep the stated limit

This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

Change one thing in the pictured case

Change one thing

Change: Use a character **vocabulary** covering b,o,k,s and space instead.

Predicted result

Prediction: The same phrase uses ten character **token**s.

Why the result follows

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Text as Pieces, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

2. State the decision to explain

Write the question to explain: Count **token**s and state the limit of the example.

3. Reveal the worked reasoning

Reveal after **prediction**s: Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

Hold back the independent task

Keep for an independent attempt: A classroom **token**izer splits sunflower into sun and flower. How many pieces result? Must every real **token**izer split it the same way?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: c at gives [2,6,3,4], four text **token**s. cat! fails on ! at zero-based position 3 after partial [1].

Exit: deleting ! changes the **input**; report unsupported punctuation or use a separately specified pre**process**ing/versioned **vocabulary**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read section 1 and complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the worked case in section 2 and complete W2.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Completion practice

Completion task: Encode cats sat using the given **vocabulary**, decode it, and count with and without an appended end marker.

Reasoning to listen for

Teacher check: Teacher explanation: Text IDs [1,5,6,7] decode cats sat. Four text **token**s; five sequence items if the protocol appends end ID 0.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Token**isation and **vocabulary**. Proposed explanation: “A **token** always equals a word”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: The **vocabulary** contains both whole strings and character pieces, so one word can take multiple **token**s.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Why is the unknown ! at position 10 rather than position 9?

Reasoned teacher answer

books occupies 0–4, space is 5, book is 6–9, so ! is character 10.

Changed-case reasoning: The same phrase uses ten character **token**s. There are five letters, one space and four letters; its decoded text remains unchanged.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Text as Pieces investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Cut sentence cards into two plausible **tokenization**s, compare piece counts, remove **context**, and predict which continuation becomes less certain.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For **vocabulary** without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A **token** always equals a word: The **vocabulary** contains both whole strings and character pieces, so one word can take multiple **token**s.

IDs measure semantic similarity: IDs are **vocabulary** indices; their numerical spacing does not establish meaning.

Dropping characters preserves an exact round trip: Deleted characters change the source and must not be hidden.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Text as Pieces questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or

letters.

Independent question

Independent question · withhold the key

A classroom **token**izer splits sunflower into sun and flower. How many pieces result? Must every real **token**izer split it the same way?

Expected independent answer

Teacher key · Expected independent answer

Two pieces result under this supplied scheme. Real **token**izers use different vocabularies and rules, so their splits may differ.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “One word always equals one **token**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

A uses 3 **token**s, B uses 1. Real **token**izers can split words, include spaces or punctuation and use different vocabularies.

A word count alone cannot determine a **token** budget; these toy splits illustrate the idea rather than document a particular model.

Explanations for the additional workbook examples

W1 Read the contract

Workbook question · W1

Explain why sat wins over s at the start of sat and why uppercase C is unsupported.

Teacher reasoning · W1

Teacher explanation: The rule chooses the longest matching ordinary **token**; sat has length three and s length one. **Vocabulary** and matching are case sensitive and contain no uppercase C.

W2 Encode and decode

Workbook question · W2

Encode cats sat using the given **vocabulary**, decode it, and count with and without an appended end marker.

Teacher reasoning · W2

Teacher explanation: Text IDs [1,5,6,7] decode cats sat. Four text **token**s; five sequence items if the protocol appends end ID 0.

W3 Compare sequence lengths

Workbook question · W3

Encode cats sat with character **token**s only. Explain the difference from four text **token**s in the original **vocabulary**.

Teacher reasoning · W3

Teacher explanation: c,a,t,s,space,s,a,t is eight text **token**s. Larger chunks in the original **vocabulary** compress this representation; the source text is unchanged.

W4 Trace a new input independently

Workbook question · W4

Under the original **vocabulary** encode sat cats, decode it and count text **token**s. Then trace sat cat! and state the exact failure position using zero-based character positions.

Teacher reasoning · W4

Teacher explanation: sat cats encodes [7,6,1,5], decodes exactly and has four text **token**s. sat cat! has partial IDs [7,6,1] then unsupported ! at position 7. Report failure with the partial trace, not complete success.

W5 Audit silent repair

Workbook question · W5

An encoder lowercases Cat and removes punctuation before **encoding**, but labels its **output** an exact **encoding** of the original. Explain what must be recorded or changed.

Teacher reasoning · W5

Teacher explanation: Lowercasing and punctuation removal alter the source.

Record pre**process**ing separately and label the transformed text; an exact encoder under this contract must report unsupported characters rather than silently change them.

Assess independent transfer

Independent question · withhold the key

Selected case: Under the original **vocabulary** encode sat cats, decode it and count text **token**s. Then trace sat cat! and state the exact failure position using zero-based character positions.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: sat cats encodes [7,6,1,5], decodes exactly and has four text **token**s. sat cat! has partial IDs [7,6,1] then unsupported ! at position 7. Report failure with the partial trace, not complete success.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Text as Pieces to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Tokenize book descriptions before any text-based matching, while preserving the original source text.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 26: Image and Voice Pipelines. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires [7,6,1,5], four text **token**s and unsupported ! at position 7. Fresh retry: cats cat encodes [1,5,6,1], four text **token**s; cats Cat fails at uppercase C position 5 after [1,5,6].

Require exact **decoding** and an explicit failure.

Week 3 uses **vocabulary** alternatives to inspect next-**token** probability distributions and sequence probabilities.

Plan the delayed return

Suggested return points in this level: ai-2/26, ai-2/28, ai-2/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Document the recommender's text representation and unknown-**token** behavior rather than assuming every model counts words identically.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Text as Pieces lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Encode and decode text under an explicit toy **vocabulary**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Encode cats sat using the given **vocabulary**, decode it, and count with and without an appended end marker.

Supported practice answer

Reasoned practice answer: Teacher explanation: Text IDs [1,5,6,7] decode cats sat. Four text **token**s; five sequence items if the protocol appends end ID 0.

Check every learner before discussion: Use the worked case for **Token**isation and **vocabulary**. Proposed explanation: “A **token** always equals a word”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: The **vocabulary** contains both whole strings and character pieces, so one word can take multiple **token**s.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Use a character **vocabulary** covering b,o,k,s and space instead. Require a **prediction** and a reason before comparing. Expected result: The same phrase uses ten character **token**s.

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

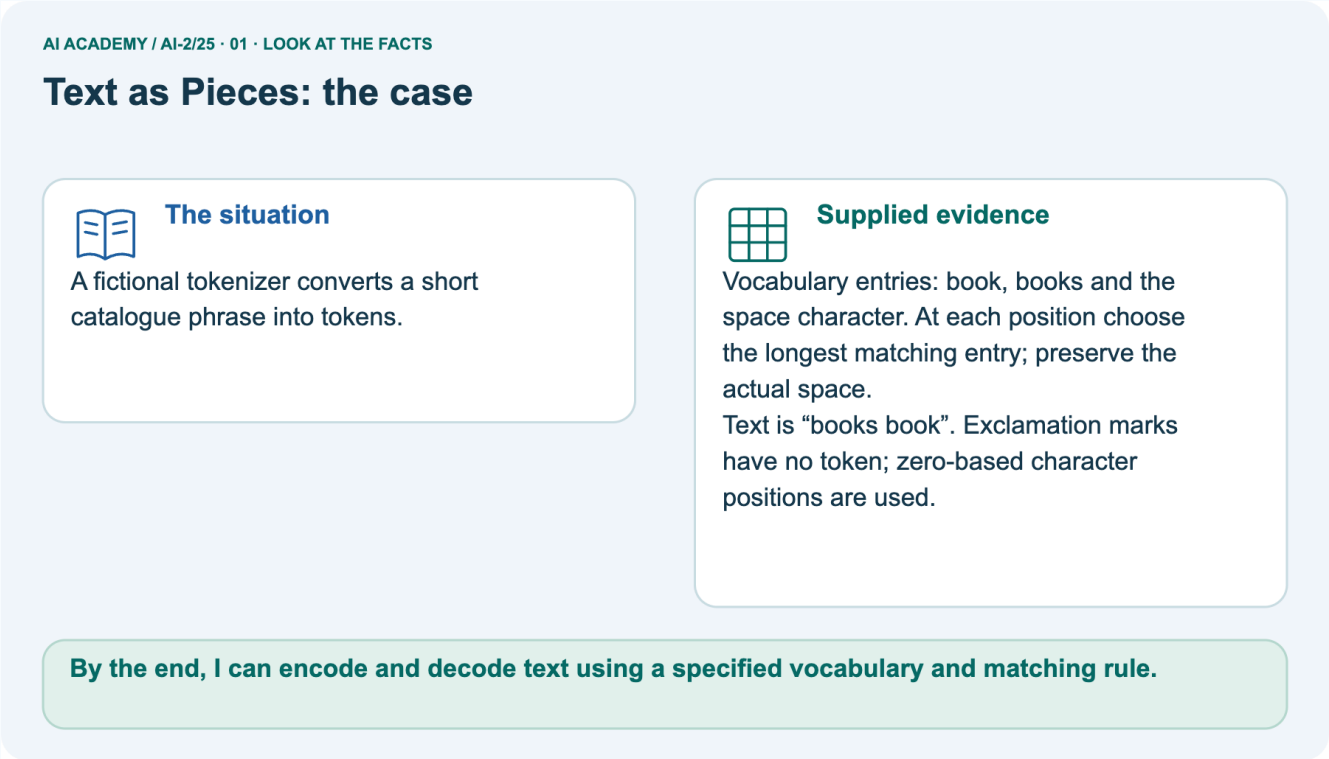
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Text as Pieces graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Text as Pieces: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional **token**izer converts a short catalogue phrase into **token**s. Supplied facts: **Vocabulary** entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

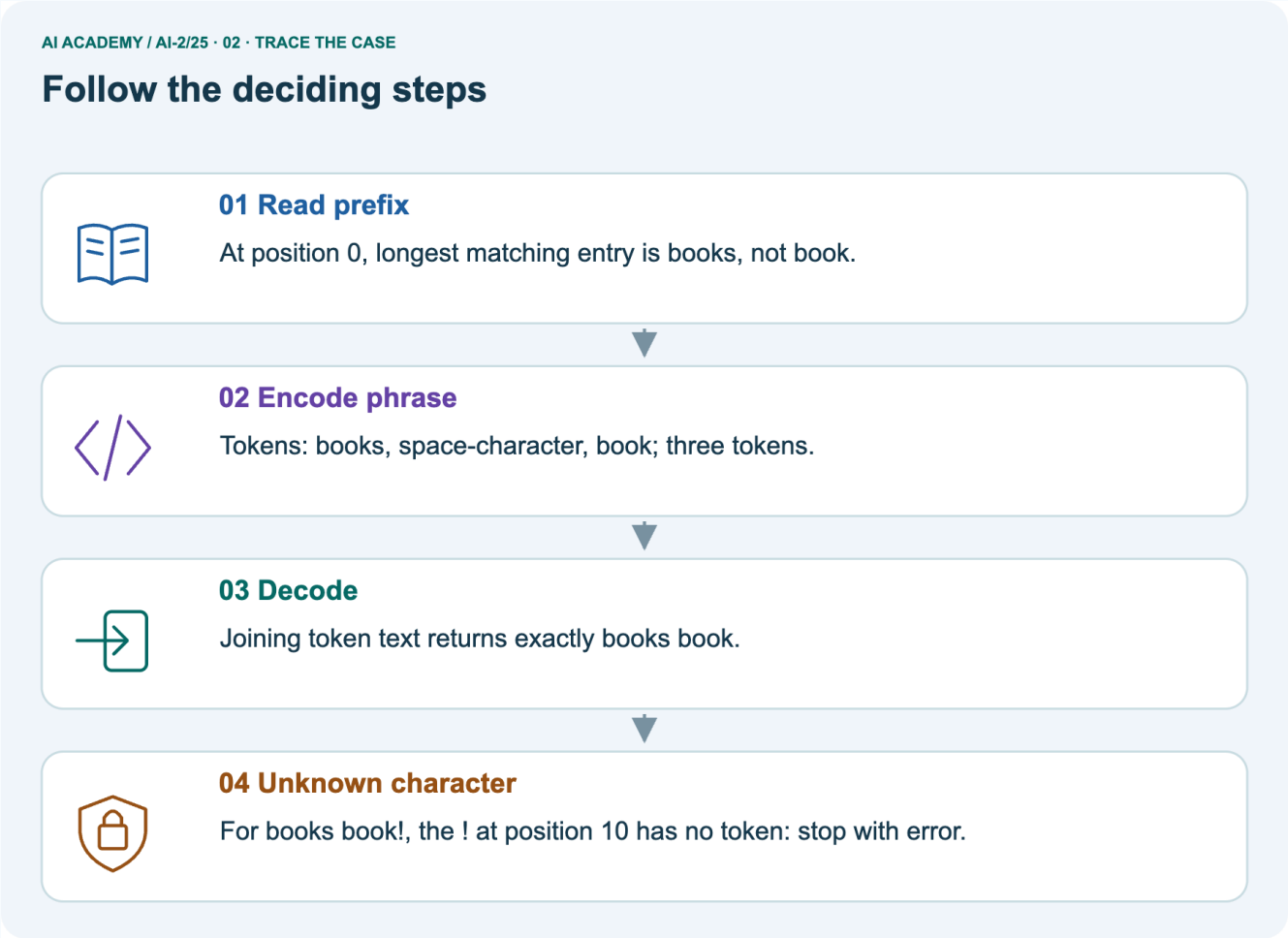
Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

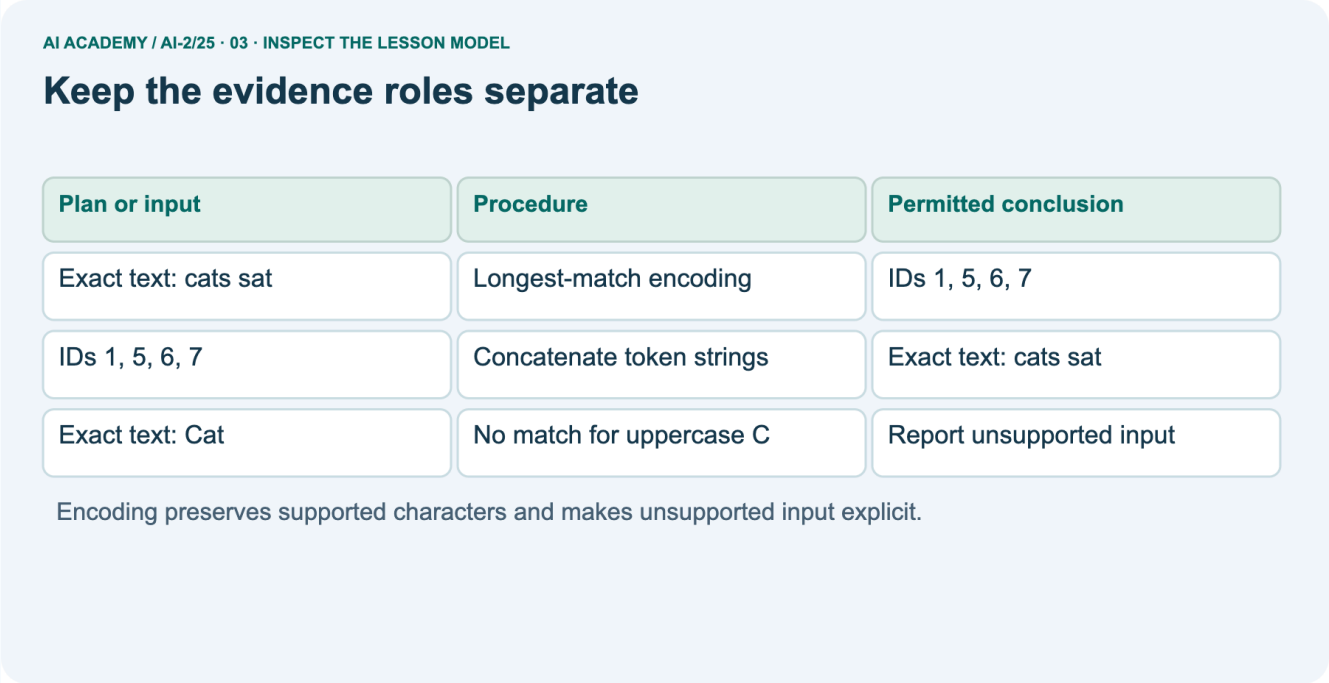
Read prefix: At position 0, longest matching entry is books, not book. → Encode phrase: **Token**s: books, space-character, book; three **token**s. → Decode: Joining **token** text returns exactly books book. → Unknown character: For books book!, the ! at position 10 has no **token**: stop with error.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. Exact text: cats sat | Longest-match **encoding** | IDs 1, 5, 6, 7. IDs 1, 5, 6, 7 | Concatenate **token** strings | Exact text: cats sat.

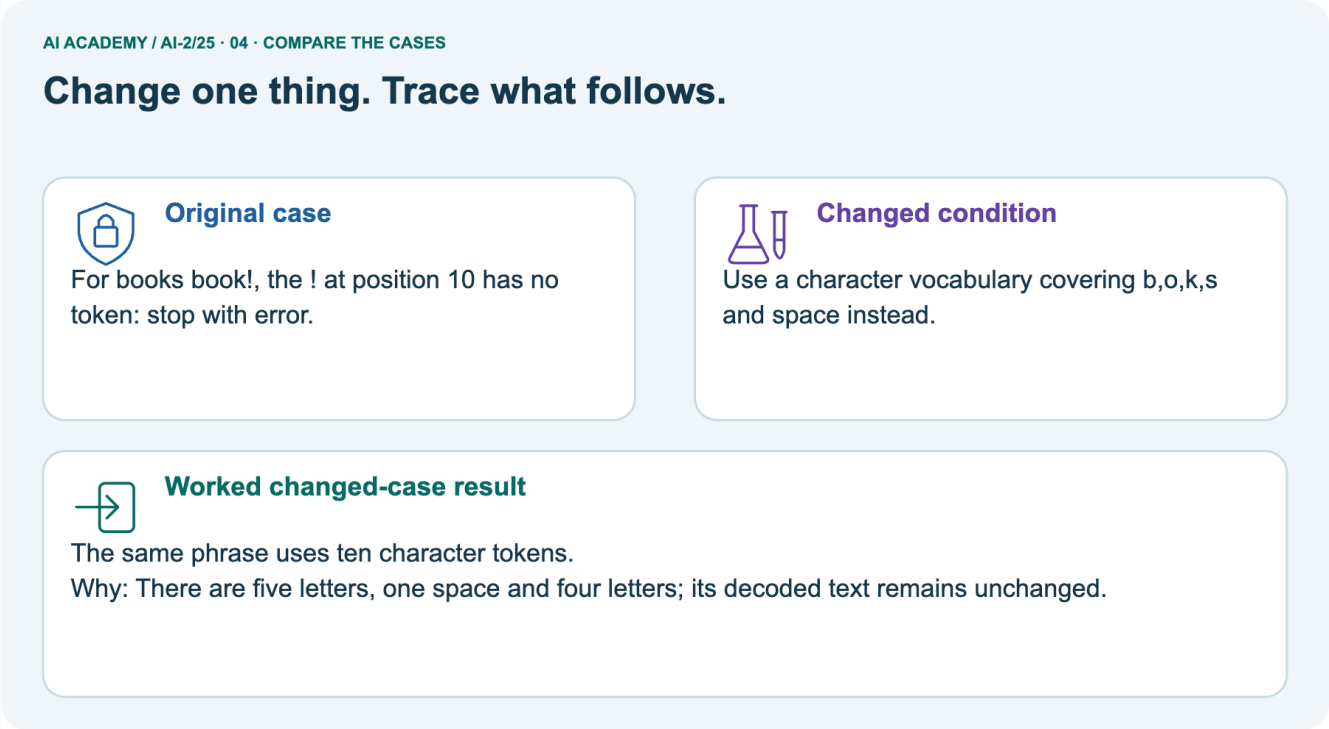
Exact text: Cat | No match for uppercase C | Report unsupported **input**

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: For books book!, the ! at position 10 has no **token**: stop with error. Changed condition: Use a character **vocabulary** covering b,o,k,s and space instead. Expected result: The same phrase uses ten character **token**s.

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

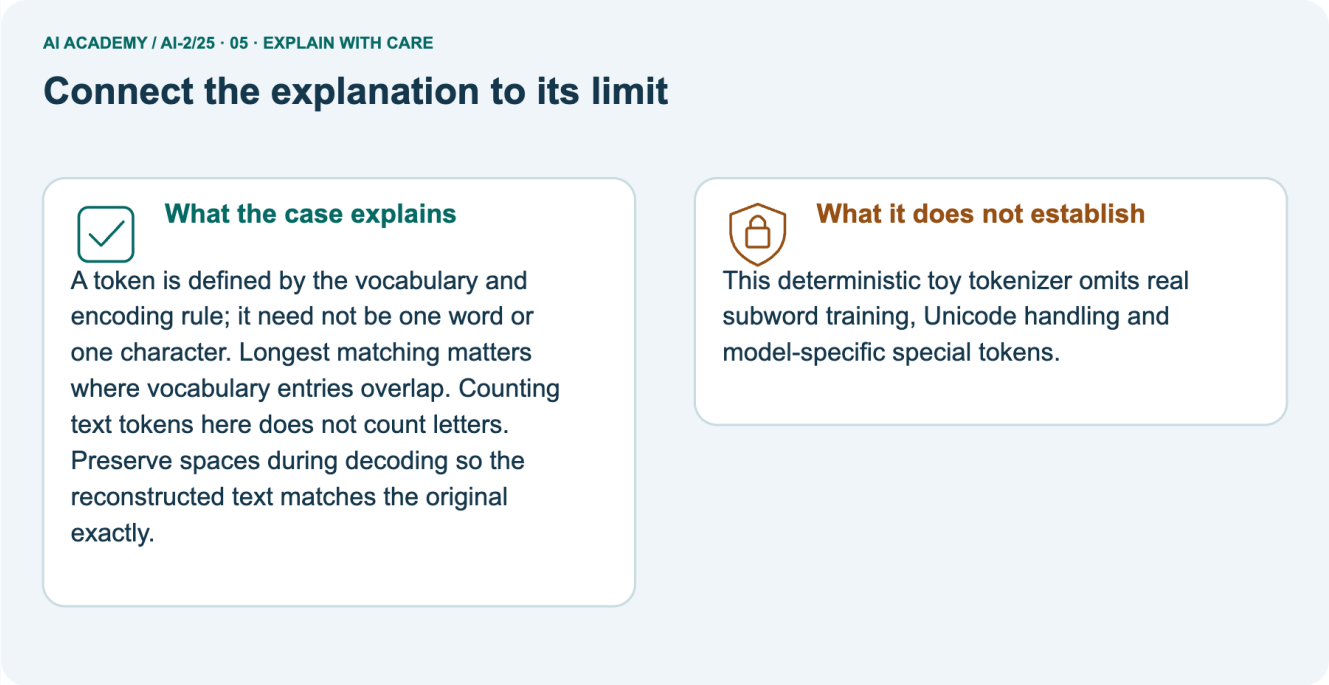
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: There are five letters, one space and four letters; its decoded text remains unchanged.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly. Limit: This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

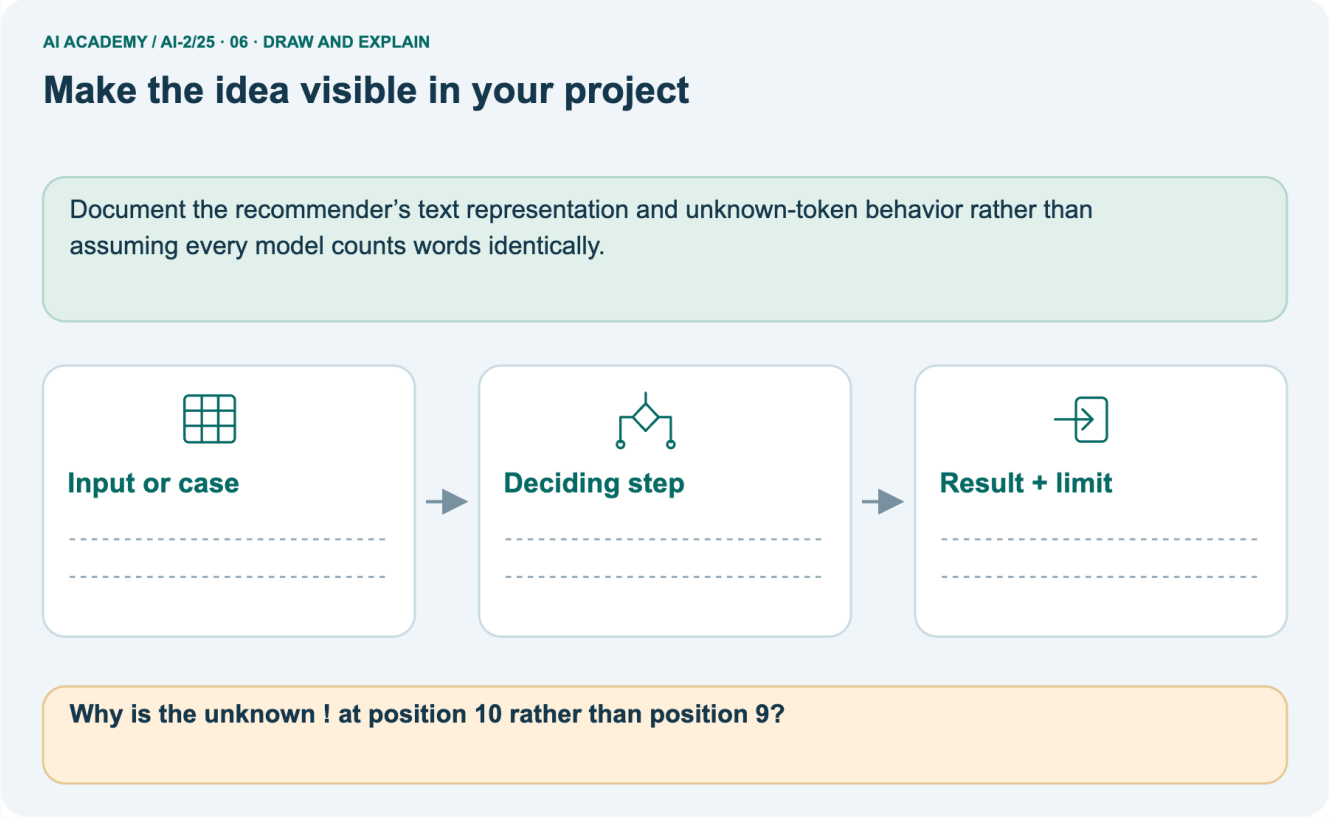
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document the recommender’s text representation and unknown-**token** behavior rather than assuming every model counts words identically. Draw your **input**, deciding step, result and limit.

Supported question: Why is the unknown ! at position 10 rather than position 9?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: books occupies 0–4, space is 5, book is 6–9, so ! is character 10.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.